

Mark Scheme

Q1.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)		B1	This mark is given for a correct shape and labels for a tree diagram
		B1	This mark is given for the correct probabilities shown
(b)	$\frac{9}{10} \times \frac{4}{5} \times \frac{2}{3}$	M1	This mark is given for a multiplication of three probabilities
	$= \frac{12}{25}$	A1	This mark is given for the correct probability that Sasha selects three marbles
(c)	$\frac{9}{10} \times \frac{1}{5} + \frac{4}{5} \times \frac{1}{3}$	M1	This mark is given for the addition of two products
	$= \frac{21}{50}$	A1	This mark is given for the correct probability that Sasha selects at least one marble of each colour
(d)	$P(\text{red from } B \mid \text{red selected}) =$ $\frac{\frac{9}{10} \times \frac{1}{5}}{1 - \frac{12}{25}} = \frac{9}{50} \times \frac{25}{13}$	M1	This mark is given for determining the correct ratio of probabilities
	$= \frac{9}{26}$	A1	This mark is given for the correct probability that Sasha selects a red marble from bag B
(Total 8 marks)			

Q2.

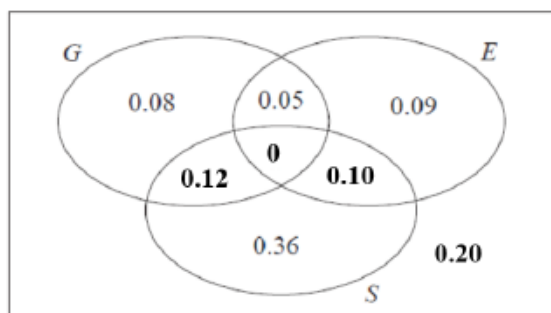
	Scheme	Marks	AO
(a)	A, C or D, B or D, C	B1 (1)	1.2
(b)	$[p = 0.4 - 0.07 - 0.24 =]$ <u>0.09</u>	B1 (1)	1.1b
(c)	A and B independent implies $P(A) \times 0.4 = 0.24$ or $(q + 0.16 + 0.24) \times 0.4 = 0.24$ so $P(A) = 0.6$ and $q =$ <u>0.20</u>	M1 A1cso (2)	1.1b
(d)(i)	$P(B' C) = 0.64$ gives $\frac{r}{r+p} = 0.64$ or $\frac{r}{r+0.09} = 0.64$ $r = 0.64r + 0.64p$ so $0.36r = 0.0576$ so $r =$ <u>0.16</u>	M1 A1	3.1a 1.1b
(ii)	Using sum of probabilities = 1 e.g. $0.6 + 0.07 + 0.25 + s = 1$ so $s =$ <u>0.08</u>	M1 A1 (4)	1.1b 1.1b
		(8 marks)	

	Notes
(a)	B1 for one correct pair. If more than one pair they must all be correct. Condone in a correct probability statement such as $P(A \cap C) = 0$ or correct use of set notation e.g. $A \cap C = \emptyset$ BUT e.g. “ $P(A)$ and $P(C)$ are mutually exclusive” alone is B0
(b)	B1 for $p = 0.09$ (Maybe stated in Venn Diagram [VD]) [If values in VD and text conflict, take text or a value <u>used</u> in a later part]
(c)	M1 for a correct equation in one variable for $P(A)$ or q using independence or for seeing both $P(A \cap B) = P(A) \times P(B)$ and $0.24 = 0.6 \times 0.4$ A1cso for $q = 0.20$ or exact equivalent (dep on correct use of independence) Beware Use of $P(A) = 1 - P(B) = 0.6$ leading to $q = 0.2$ scores M0A0
(d)(i)	1 st M1 for use of $P(B' C) = 0.64$ leading to a correct equation in r and possibly p . Can fit their p provided $0 < p < 1$ 1 st A1 for $r = 0.16$ or exact equivalent
(ii)	2 nd M1 for use of total probability = 1 to form a linear equation in s . Allow p, q, r etc Can follow through their values provided each of p, q, r are in $[0, 1)$ 2 nd A1 for $s = 0.08$ or exact equivalent

Q3.

	Scheme	Marks	AO
(a)	$0.08 + 0.09 + 0.36 = \underline{0.53}$	B1 (1)	1.1b
(b)(i)	$[P(G \cap E \cap S) = 0 \Rightarrow] \underline{p = 0}$	B1	1.1b
(ii)	$[P(G) = 0.25 \Rightarrow] 0.08 + 0.05 + q + "p" = 0.25$ $\underline{q = 0.12}$	M1 A1 (3)	1.1b 1.1b
(c)(i)	$\left[P(S E) = \frac{5}{12} \Rightarrow \right] \frac{r + "p"}{r + "p" + 0.09 + 0.05} = \frac{5}{12}$ $[12r = 5r + 5 \times 0.14 \Rightarrow] \underline{r = 0.10}$	M1 A1ft A1	3.1a 1.1b 1.1b
(ii)	$[0.08 + 0.05 + "0.12" + "0" + 0.09 + "0.10" + 0.36 + t = 1 \Rightarrow] \underline{t = 0.20}$	B1ft (4)	1.1b
(d)	$P(S \cap E') = 0.36 + "q" [= 0.48]$ $P([(S \cap E')] \cap G) = "q" [= 0.12] \text{ and } P(G) = 0.25 \text{ and}$ $P(S \cap E') \times P(G) = "0.48" \times \frac{1}{4} \text{ or } 0.12$ $P(S \cap E') \times P(G) = 0.12 = P([(S \cap E')] \cap G) \text{ so are independent}$	B1ft M1 A1 (3)	1.1b 2.1 2.2a
		(11 marks)	

	Notes
(a)	B1 for 0.53 (or exact equivalent) [Allow 53%]
(b)(i)	B1 for $p = 0$ (may be placed in Venn diagram)
(ii)	M1 for a linear equation for q (ft letter " p " or their value if $0 \leq p \leq 0.12$) $\Rightarrow$ by $p + q = 0.12$ A1 for $q = 0.12$ (may be placed in Venn diagram)
(c)(i)	M1 for a ratio of probabilities (r on num and den) (on LHS) with num < den and num or den correct ft. Allow ft of letter " p " or their p where $0 \leq p < 0.86$ but "+ 0" is not required. 1 st A1ft for a correct ratio of probabilities (on LHS) allowing ft of their p where $0 \leq p < 0.86$ 2 nd A1 for $r = 0.1(0)$ or exact equivalent (may be in Venn diagram) Ans only 3/3
(ii)	B1ft for $t = 0.2(0)$ (o.e.) or correct ft i.e. $0.42 - (p + q + r)$ where p, q, r and t are all probs
(d)	B1ft for $P(S \cap E') = 0.48$ (with label) (ft letter " q " or their value if $0 \leq q \leq 0.12$) M1 for attempting all required probs (labelled) and using them in a correct test (allow ft of q) A1 for all probs correct and a correct deduction (no ft deduction here) SC No "P" If correct argument seen apart from P for probability for all 3 marks, award (B0M1A1) If unsure about an attempt using conditional probabilities, please send to review.



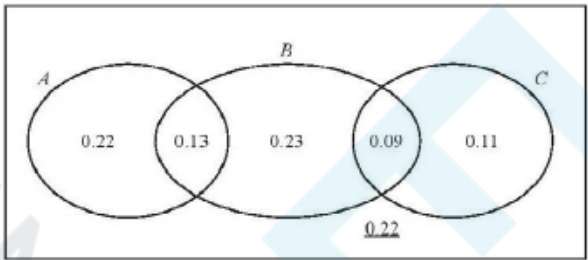
Q4.

	Scheme	Marks	AO
(a)	Disadvantage: e.g. Not random; cannot use (reliably) for inferences	B1	1.1b
(b)	[Sight or correct use of] $X \sim B(36, 0.08)$	M1	3.3
(i)	$P(X=4) = 0.167387\dots$ awrt <u>0.167</u>	A1	1.1b
(ii)	$[P(X \leq 7) = 1 - P(X \geq 6)]$ 0.022233... awrt <u>0.0222</u>	A1	1.1b
(c)	$P(\text{In dance club and dance tango}) = 0.4 \times 0.08 = \underline{0.032}$ or $\frac{4}{125}$ or <u>3.2%</u>	B1	1.1b
(d)	[Let T = those who can dance the Tango. Sight or use of] $T \sim B(50, "0.032")$ $[P(T < 3) = P(T \leq 2)]$ 0.7850815... awrt <u>0.785</u>	M1 A1	3.3 1.1b
		(2)	
		(7 marks)	
	Notes		
(a)	B1 for a suitable disadvantage:		
	Allow (B1)	Do NOT allow (B0)	
	Not random <u>or</u> less random (o.e.)	Not representative	
	Cannot use (reliably) for inferences	Less accurate	
	(More likely to be) biased	Any comment based on time or cost	
		Any mention of skew	
		Any mention of non-response	
(b)	M1 for sight of $B(36, 0.08)$ Allow in words: <u>binomial</u> with $n = 36$ and $p = 0.08$ may be implied by one correct answer to 2sf <u>or</u> sight of $P(X \leq 6) = 0.97776\dots$ i.e. awrt 0.98 Allow for $36C4 \times 0.08^4 \times 0.92^{32}$ as this is "correct use"		
(i)	1 st A1 for awrt 0.167 NB An answer of just awrt 0.167 scores M1($\Rightarrow$) 1 st A1		
(ii)	2 nd A1 for awrt 0.0222		
(c)	B1 for 0.032 o.e. (Can allow for sight of 0.4×0.08)		
(d)	M1 for sight of $B(50, "0.032")$ ft their answer to (c) provided it is a probability $\neq 0.08$ may be implied by correct answer <u>or</u> sight of $[P(T \leq 3)] = 0.924348\dots$ i.e. awrt 0.924 or $P(T \leq 2)$ as part of $1 - P(T \leq 2)$ calc.		
	A1 for awrt 0.785		
MR	Allow MR of 50 (e.g. 30) provided clearly attempting $P(T \leq 2)$ and score M1A0		

Q5.

	Scheme	Marks	AO
(a)	{Let X = time spent, $P(X > 15) = \dots$ awrt <u>0.106</u>	B1 (1)	1.1b
(b)	$H_0: \mu = 10$ $H_1: \mu > 10$ $\bar{X} \sim N\left(10, \left(\frac{4}{\sqrt{20}}\right)^2\right)$; $P(\bar{X} > 11.5) = 0.046766\dots$ [Condone 0.9532...] [This is significant (< 5%) so] there is evidence to support the complaint	B1 M1;A1 A1 (4)	2.5 3.3;3.4 2.2b
(c)(i)	$P(T < 2) = \dots$ awrt <u>0.196</u>	B1 (1)	1.1b
(ii)	Require $\frac{P(0 < T < 2)}{P(T > 0)} = \frac{0.119119\dots}{0.923436\dots}$; = 0.1289955... awrt <u>0.129</u>	M1 A1;A1 (3)	3.4 1.1bx2
(iii)	The current model suggests non-negligible probability of T values < 0 which is impossible	B1 (1)	3.5b
(d)	Require t such that $P(T > t T > 2) = 0.5$ or $P(T < t T > 2) = 0.5$ e.g. $\frac{P(T > t)}{P(T > 2)} = 0.5$; so $P(T > t) = 0.5 \times [1 - (c)(i)]$ or $P(T > t) = 0.5 \times 0.8043\dots$ [i.e. $P(T > t) = 0.40\dots$ implies] $\frac{t-5}{3.5} = 0.2533$ or $P(T < t) = "0.5978\dots"$ $t = 5.886\dots$ or from calculator 5.867... so awrt <u>5.9</u>	M1 M1; A1ft M1 A1 (5)	3.1b 1.1b 3.4 1.1b 1.1b
		(15 marks)	

	Notes
(a)	B1 for awrt 0.106 (from calculator) [Allow 10.6%]
(b)	B1 for both hypotheses correct in terms of μ M1 for selection of a correct model (sight or use of correct normal- may not have label $\bar{X}$) 1 st A1 for use of this model to get probability allow 0.046~0.047 [Condone awrt 0.953] ALT OR test statistic $z = 1.677\dots$ (awrt 1.68) and cv of 1.64 (or better) or CR $\bar{X} > 11.47\dots$ 2 nd A1 (dep on 1 st A1 or at least $P(\bar{X} > 11.5) < 0.05$ (o.e.)) for a correct conclusion in context -must mention complaint/claim or time/mins is > 10 SC (M0 for $\bar{X} \sim N(11.5, \dots)$ for correct probability and conclusion (score M0A0A1 on open)
(c)(i)	B1 for awrt 0.196 (from calculator) [Allow 19.6%]
(ii)	M1 for a correct probability ratio expression (may be implied by 1 st A1 scored) 1 st A1 for a correct ratio of probabilities (both correct or truncated to 2 dp) 2 nd A1 for awrt 0.129
(iii)	B1 for a suitable explanation of why model is not suitable based on negative T values Must say that a significant proportion of values < 0 (o.e.) e.g. $P(T > 0)$ should be closer to 1 or Difference between $P(T < 2 T > 0)$ and $P(T < 2)$ is too big (o.e.)
(d)	1 st M1 for a correct conditional probability statement to start the problem or $0.5 \times P(T > 2)$ 2 nd M1 for correct ratio of probability expressions [Must have $P(T > t)$ or $P(2 < T < t)$] 1 st A1ft for a correct equation for $P(T > t)$ (o.e.) ft their answer to part (c) [May be in a diagram] 3 rd M1 for attempt to find t (standardising and sight of 0.2533) or prepare to use calc (ft) Arriving at $P(T < \text{median}) = 1 - 0.5 \times \text{"their 0.8043"}$ will score 1 st 4 marks 2 nd A1 for awrt 5.9 Sight of awrt 5.9 and at least one M mark scores 5/5 [Answer only send to review]

Question	Scheme	Marks	AOs
(a)	$P(A' B') = \frac{P(A' \cap B')}{P(B')} \text{ or } \frac{0.33}{0.55}$	M1	3.1a
	$= \frac{3}{5} \text{ or } 0.6$	A1	1.1b
		(2)	
(b)	e.g. $P(A) \times P(B) = \frac{7}{20} \times \frac{9}{20} = \frac{63}{400} \neq P(A \cap B) = 0.13 = \frac{52}{400}$ or $P(A' B') = 0.6 \neq P(A') = 0.65$	B1	2.4
		(1)	
(c)		B1	2.5
		M1	3.1a
		A1	1.1b
		M1	1.1b
		A1	1.1b
(d)	$P(B \cup C)' = 0.22 + 0.22 \text{ or } 1 - [0.56]$ or $1 - [0.13 + 0.23 + 0.09 + 0.11]$	M1	1.1b
	$= 0.44$	A1	1.1b
		(2)	
(10 marks)			

Notes:

(a)

M1: for a correct ratio of probabilities formula and at least one correct value.**A1:** a correct answer

(b)

for a fully correct explanation: correct probabilities and correct comparisons.

(c)

B1: for box with B intersecting A and C but C not intersecting A . (Or accept three intersecting circles, but with zeros entered for $A \cap C$ and $A \cap B \cap C$) No box is B0**M1:** for method for finding $P(B \cap C)$ **A1:** for 0.09**M1:** for 0.13 and their 0.09 in correct places and method for their 0.23**A1:** fully correct

(d)

M1: for a correct expression – fit their probabilities from their Venn diagram.**A1:** cao

Q7.

Qu	Scheme	Marks	AO
(a)	$P(L > 16) = 0.69146...$ awrt 0.691	B1 (1)	1.1b
(b)	$P(L > 20 L > 16) = \frac{P(L > 20)}{P(L > 16)}$ $= \frac{0.308537...}{(a)} \text{ or } \frac{1-(a)}{(a)} = 0.44621...$ For calc to work require $(0.44621...) ^4 = 0.03964...$ awrt <u>0.0396</u>	M1 A1ft, A1 dM1 A1 (5)	3.1b 1.1b 1.1b 2.1 1.1b
(c)	Require: $[P(L > 4)]^2 \times [P(L > 20 L > 16)]^2$ $= (0.99976...) ^2 \times ("0.44621...")^2$ $= 0.19901...$ awrt <u>0.199</u> (*)	M1 A1ft A1cso* (3)	1.1a 1.1b 1.1b
(d)	$H_0 : \mu = 18 \quad H_1 : \mu > 18$ $\bar{L} \sim N\left(18, \left(\frac{4}{\sqrt{20}}\right)^2\right)$ $P(\bar{L} > 19.2) = P(Z > 1.3416...) = 0.089856...$ (0.0899 > 5%) or (19.2 < 19.5) or 1.34 < 1.6449 so not significant Insufficient evidence to support Alice's claim (or belief)	B1 M1 A1 A1 A1 (5)	2.5 3.3 3.4 1.1b 3.5a
(14 marks)			

Notes	
(a)	B1 for evaluating probability using their calculator (awrt 0.691) Accept 0.6915
(b)	1 st M1 for a first step of identifying a suitable conditional probability (either form) 1 st A1ft for a ratio of probabilities with numerator = awrt 0.309 or 1 – (a) and denom = their (a) 2 nd A1 for awrt 0.446 (o.e.) Accept 0.4465 (from $\frac{0.3085}{0.691} = 0.44645...$) NB $\frac{P(16 < L < 20)}{P(L > 16)} = 0.5538...$ scores M1A1A1 when they do $1 - 0.5538 = 0.4462...$ 2 nd M1 (dep on 1 st M1) for 2 nd correct step i.e. (their 0.446...) ⁴ or $X \sim B(4, "0.446")$ and $P(X = 4)$ 3 rd A1 for awrt 0.0396
(c)	1 st M1 for a correct approach to solving the problem (May be implied by A1ft) 1 st A1ft for $P(L > 4) =$ awrt 0.9998 used and ft their 0.44621 in correct expression If use $P(L > 20) = 0.3085...$ as 0.446.. in (b) then M1 for $(0.3085...) ^2 \times [P(L > 4)]^2$; A1ft as above * 2 nd A1cso for 0.199 or better with clear evidence of M1 [NB $(0.4662...) ^2 = 0.199...$ is M0A0A0] Must see M1 scored by correct expression in symbols or values (M1A1ft)
(d)	B1 for both hypotheses in terms of μ . M1 for selecting a suitable model. Sight of <u>normal</u> , <u>mean</u> 18, <u>sd</u> $\frac{4}{\sqrt{20}}$ (o.e.) or <u>variance</u> = 0.8 1 st A1 for using the model correctly. Allow awrt 0.0899 or 0.09 from correct prob. statement ALT CR $(\bar{L}) > 19.471...$ (accept awrt 19.5) or CV of 1.6449 (or better: calc 1.6448536..)
	2 nd A1 for correct non-contextual conclusion. Wrong comparison or contradictions A0 Error giving 2 nd A0 implies 3 rd A0 but just a correct contextual conclusion can score A1A1 3 rd A1 dep on M1 and 1 st A1 for a correct contextual conclusion mentioning <u>Alice's claim</u> / <u>belief</u> or there is insufficient evidence that the mean <u>lifetime</u> is more than 18 hours